

Credit Card Fraud Detection: A Comparative Study of Classical ML, Deep Learning, Generative Oversampling, Ensembles and Graph Neural Networks

With a self-contained introduction to machine-learning model families

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Abstract

We build and rigorously evaluate a complete fraud-detection pipeline on the public Kaggle ULB credit-card dataset (284,807 transactions, 492 frauds, 0.173% prevalence). Seventy-two experiments span five model families — linear models, tree ensembles, neural networks, generative-augmentation pipelines, and graph neural networks — combined with seven imbalance-handling strategies, all under one deployment-honest protocol: a chronological 70/15/15 train/validation/test split, validation-only tuning, and a test set that keeps its natural imbalance. The champion, a Random Forest trained on SMOTE-ENN-resampled data, reaches $\text{PR-AUC} = 0.795$ ($F_1 = 0.848$, $\text{MCC} = 0.855$, one false positive in 42,722 transactions), the top of the honest-protocol range reported in recent literature. Nine controlled ablations explain *why*: hybrid sampling wins for trees; naive extreme class weighting silently collapses boosted models; cross-family ensembling helps while seed averaging does not; depth does not help on PCA features; and shuffling the split inflates scores by up to +0.06, reproducing the numbers that dominate older literature. Stratified bootstrap analysis shows all models are statistical ties at this test size — the contribution is the protocol and the analysis, not a leaderboard winner. Every prediction is explained with SHAP, whose feature attributions independently confirm the exploratory analysis. The report doubles as a self-contained tutorial: each model family is introduced from first principles before its experimental results are discussed.

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1 Introduction

Card-not-present fraud costs the payments industry tens of billions of dollars annually, and every fraudulent transaction that slips through is a direct loss while every legitimate transaction that is wrongly blocked damages customer trust. Machine-learning classifiers sit at the heart of modern fraud engines: they score each incoming transaction, and a policy layer decides whether to approve, challenge, or block.

This project builds such a scoring system end-to-end and asks three questions:

1. **Which model family** detects fraud best when the evaluation is done honestly?
2. **How should extreme class imbalance** (one fraud per 578 legitimate transactions) be handled — resampling, cost re-weighting, synthetic generation, or anomaly detection?
3. **How much do methodology choices** (split protocol, tuning discipline, uncertainty reporting) change the conclusions?

The project is also written as a **learning document**. Section 4 introduces every model family from first principles so that a reader new to machine learning can follow exactly what was trained and why. All 72 experiments, including failures, are logged in the accompanying experiment inventory, and every design decision is backed by a controlled ablation rather than a citation.

1.1 Reference studies: what they do, what they report, where we land

Before any result is presented, Table 1 fixes the comparison set. Every figure below was read from the cited paper, and each is paired with the number from this study that is actually comparable to it — matching *protocol* first, model second, because Section 7.6 shows protocol dominates.

Table 1: Verified published reference points and the comparable result from this work. PR-AUC throughout. The only protocol-matched row — chronological versus chronological — is the first.

Study	Protocol / data	Theirs	Ours	Δ
Sang 2026 [5]	ULB, chronological	0.7902	0.7947	+0.005
Sang 2026 [5]	ULB, random stratified	0.8815	0.840	−0.042
IJSSE 2025 [22]	ULB, random (RF / XGB)	0.871 / 0.867	0.812 / 0.810	−0.059
Anand & Chakrabarty 2025 [3]	ULB, random (LSTM ensemble)	0.8269	0.796	−0.031
HG-AE [7]	<i>relational</i> graph, entity IDs	≈0.89	0.739	n/a
KAIS 2026 [4]	private bank data, random CV	0.9879	—	n/a

The closest study is Sang [5], which evaluates ULB under *both* protocols and is therefore the only fair benchmark for this work. Table 2 sets out what it does and where this study diverges.

Read together: this study reproduces that paper’s protocol and adds breadth (five model families, a GNN tier, seven imbalance strategies) plus uncertainty quantification, while that paper adds an amount-aware cost objective this study does not implement.

2 Background: the minimum machine learning needed here

2.1 Supervised binary classification

In **supervised learning** we have labelled examples $(\mathbf{x}_i, y_i)$ where $\mathbf{x}_i$ is a feature vector and $y_i \in \{0, 1\}$ the label (here: 1 = fraud). A model f maps $\mathbf{x}$ to either a hard decision or, more usefully, a **score** $\hat{p}(y=1 \mid \mathbf{x}) \in [0, 1]$. Training means choosing parameters to minimise a loss; the

Table 2: Techniques: the protocol-matched study versus this work.

Technique	Sang 2026 [5]	This work
Chronological split	yes (66/14/20)	yes (70/15/15)
PR-AUC as primary metric	yes	yes
Threshold tuned on validation	yes (F_1 + amount-aware)	yes (F_1)
Optuna hyperparameter search	yes	yes (30 trials default)
Cost-sensitive class weighting	yes	yes (validation-selected grid)
Synthetic resampling (SMOTE/GAN)	no, by design	yes — tested as an ablation
SHAP explainability	yes	yes
Amount-aware cost threshold	yes	no (future work)
Bootstrap confidence intervals	no	yes (1,000 replicates)
Model families compared	2 (XGBoost, MLP)	18 across 5 families
Graph neural network tier	no	yes (GraphSAGE, GATv2)

standard choice for binary problems is **binary cross-entropy**, which penalises confident wrong answers heavily.

A score only becomes a decision after comparing it to a **threshold** τ : predict “fraud” iff $\hat{p} \geq \tau$. Choosing τ is itself a modelling decision — in this project it is always tuned on a validation set (Section 6), never on test data.

2.2 The class-imbalance problem

With 99.83% negatives, a model that outputs “legitimate” for everything achieves 99.83% accuracy and catches zero fraud. Accuracy is therefore useless here. Worse, imbalance distorts training: the loss is dominated by the majority class unless counter-measures are taken (Section 5). The community-standard metric for this regime is the **Precision–Recall curve** and the area under it (PR-AUC / average precision).

2.3 Splits, validation, and leakage

Models must be evaluated on data they did not see. The subtle part is *how* the data is divided:

- A **random (shuffled) split** mixes transactions from the whole period into both sides. Patterns from the future leak backwards into training. Scores look better than reality; we quantify this inflation ourselves in Section 7.6.
- A **chronological (time-based) split** trains on the earliest transactions and tests on the latest — exactly how a deployed system experiences the world.
- A **validation set** is a third partition used for *every adaptive choice*: hyper-parameter search, class-weight selection, early stopping, decision thresholds. The test set must be opened once, after all choices are frozen. Skipping validation and tuning on test is one of the most common flaws in applied fraud papers.

3 Dataset and exploratory analysis

Source. The ULB (Université Libre de Bruxelles) credit-card dataset, released with Dal Pozzolo et al.’s research collaboration with Worldline, contains real European card transactions from September 2013 [1]. To protect cardholders, the original 28 features were transformed with principal component analysis (PCA); we call them $V_1, \dots, V_{28}$. Two interpretable columns survive: **Time** (seconds since the first transaction) and **Amount** (EUR). The label is **Class** $\in \{0, 1\}$.

Engineered features. We add three derived columns used by all models: $\log(1 + \text{Amount})$ (amounts are extremely right-skewed), and cyclic encodings of the hour of day, $\sin(2\pi h/24)$ and $\cos(2\pi h/24)$, so that hour 23 and hour 0 are close.

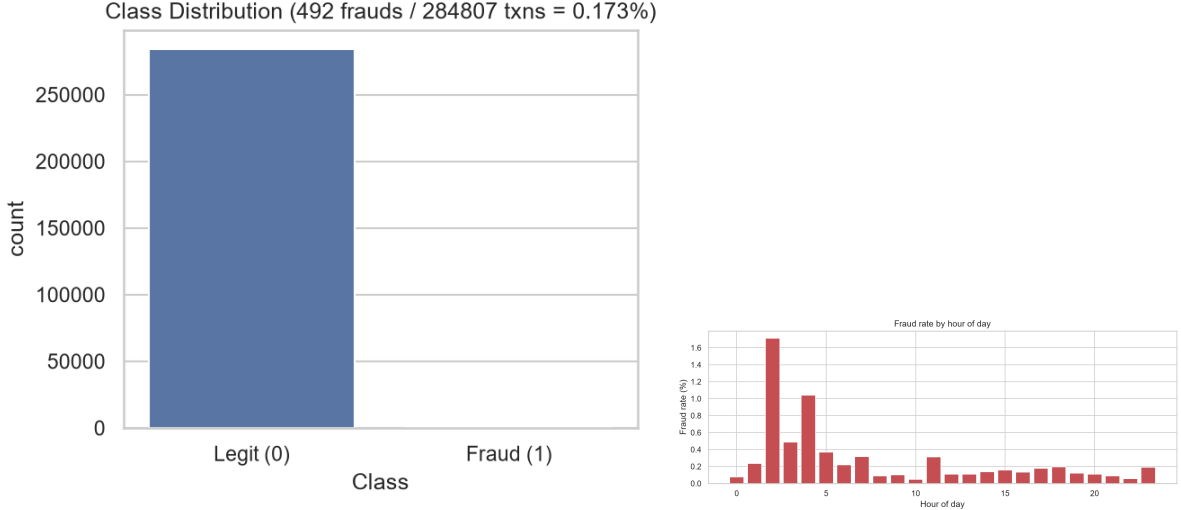


Figure 1: Left: the 578:1 class imbalance — the central difficulty of the task. **Right:** fraud rate by hour peaks around 02:00–04:00 at up to ten times the daily average, justifying time features and a chronological split.

Key exploratory findings. (i) No missing values anywhere. (ii) Fraud amounts skew small (median EUR 9.25 vs. EUR 22) yet have a heavier upper tail relative to their count — motivating the log transform. (iii) The features most correlated with the label are $V_{17}, V_{14}, V_{12}, V_{10}$ (negatively) and V_{11} (positively); no single feature separates the classes, which leaves room for non-linear models. These observations become the ground truth against which the SHAP explanations of Section 8 are later compared.

4 A taxonomy of model families

This section introduces every family used in the study, building from simplest to most complex. Table 3 at the end summarises where each of our 18 trained architectures belongs.

4.1 Linear models

Logistic Regression fits $\hat{p} = \sigma(\mathbf{w}^\top \mathbf{x} + b)$ where σ is the logistic function: a single weighted vote over features. It is fast, interpretable, and surprisingly hard to beat when signal is mostly additive. **Linear SVM** draws the maximum-margin separating hyperplane instead of maximising likelihood; empirically it lands near logistic regression here (PRAUC 0.747 vs 0.738). Both define the *linear ceiling* of the dataset.

4.2 Probabilistic baseline: Naive Bayes

Gaussian Naive Bayes applies Bayes’ rule assuming features are independent given the class. PCA components are uncorrelated *linearly*, but independence still fails badly: NB scores only ≈ 0.08 PR-AUC, a reminder that a strong assumption can be worse than no model when it is violated.

4.3 Decision trees

A **decision tree** recursively partitions feature space with axis-aligned questions (“is $V_{14} < c$?”). A single unpruned tree overfits noise: ours reaches only 0.405 PR-AUC. Trees are weak alone but are the building blocks of the strongest classical families below.

4.4 Ensemble methods: many weak learners, one strong committee

Bagging (bootstrap aggregation) trains many deep trees on random resamples and averages them — variance cancels out. **Random Forest** [11] adds extra randomness by restricting each split to a random feature subset; it is our overall champion.

Gradient boosting instead builds trees *sequentially*, each new tree fitting the residual errors of the current ensemble. Modern implementations — **XGBoost** [12], **LightGBM** [13] (histogram-based splits), and **CatBoost** [14] (ordered boosting, robust handling of categorical variables) — dominate tabular benchmarks. Their weakness discovered here: blindly setting the positive-class weight to the imbalance ratio (519) makes boosting predict almost everything as fraud (Section 7.5).

Stacking / voting combines *different* families: our best deep ensemble averages the probabilities of a forest, an MLP and an LSTM (0.785). Diversity across families is what pays off; averaging five LSTMs that differ only in random seed does not (0.775).

4.5 Neural networks

All neural nets share the same atom: a **layer** computes $\mathbf{h}' = \phi(W\mathbf{h} + \mathbf{b})$ with a non-linearity ϕ (ReLU etc.). Stacking layers lets the network compose simple features into complex ones.

MLP / DNN (feed-forward).

Fully connected layers read one transaction at a time. Our MLP (128-64-32 units) scores 0.775; doubling depth and width (DNN, 512-256-128-64 with batch normalisation) changes nothing (0.7752) — depth buys nothing on 32 PCA features.

Autoencoder.

An encoder compresses legitimate transactions to a 16-dimensional code and a decoder reconstructs them; frauds reconstruct poorly, so reconstruction error becomes an anomaly score. Trained without labels it reaches ROC-AUC 0.94 but only 0.584 PR-AUC: ignoring labels costs ranking precision exactly where fraud systems need it.

RNN / LSTM.

Recurrent networks read *sequences*, carrying a hidden state across steps; the LSTM’s gating cells remember long-range patterns. Feeding sliding windows of 8 consecutive transactions lifts performance to 0.780 — temporal context is real signal. Making it bidirectional (BiLSTM, reading forwards and backwards) adds nothing (0.781): fraud evidence is causal, not symmetric.

Transformers.

Replace recurrence with **self-attention**, letting every element weigh every other element in parallel [15]. **FT-Transformer** applies this to tabular rows (each feature becomes a token); at 0.681 it loses to boosting and even to the plain MLP — attention overhead is unjustified on 32 numeric features, consistent with recent tabular-DL studies.

GANs.

Generative Adversarial Networks [16] pit a Generator (creates fake samples from noise) against a Discriminator (real vs. fake) until fakes become statistically indistinguishable. We train one on the minority class and use it to synthesise frauds (Section 5).

4.6 Graph neural networks

When entities are linked (card—merchant—device), data forms a **graph** $G = (V, E)$, and **message passing** updates each node from its neighbours: $\mathbf{h}'_v = \text{UPDATE}(\mathbf{h}_v, \text{AGG}\{\mathbf{h}_u : u \in \mathcal{N}(v)\})$. **GraphSAGE** [17] samples a fixed number of neighbours and aggregates (mean/max/pool); **GAT** [18] learns attention weights over neighbours. Crucially, ULB has *no entity identifiers*, so our graph connects each transaction to its $k=10$ nearest neighbours in feature space — a similarity surrogate. SAGE still reaches 0.756, but attention (GAT, 0.745) adds nothing: there is no true relational structure to attend over. Optuna’s choice of a *single* SAGE layer confirms over-smoothing — two rounds of averaging near-identical neighbours blurs discriminative detail.

Table 3: Where every trained architecture sits in the taxonomy.

Family	Model	One-line idea	Best PRAUC
Linear	LogReg / LinearSVC	weighted vote / max-margin hyperplane	0.738 / 0.747
Probabilistic	GaussianNB	Bayes rule + independence assumption	0.079
Tree	DecisionTree	recursive if-else partitioning	0.405
Bagging	RandomForest	averaged de-correlated trees	0.795
Boosting	XGBoost / LightGBM / CatBoost	sequential residual fitting	0.777 / 0.775 / 0.783
Feed-forward NN	MLP / DNN	composed dense layers	0.775 / 0.775
Autoencoder	AE	reconstruction-error anomaly score	0.584
Recurrent NN	LSTM / BiLSTM	gated state over 8-step windows	0.780 / 0.781
Transformer	FT-Transformer	self-attention over feature tokens	0.681
GAN augmentation	GAN + any model	adversarially generated minority samples	0.783 (CatB)
Graph NN	GraphSAGE / GAT	neighbour message passing	0.756 / 0.745
Stacked ensemble	RF+MLP+LSTM	average of diverse families	0.785

5 Handling class imbalance

Seven strategies were compared, all applied to the *training partition only*:

1. **raw** — do nothing; establishes each model’s floor.
2. **class weighting** — multiply the positive class’s loss contribution by a weight w . The naive choice $w = n_{\text{neg}}/n_{\text{pos}} \approx 519$ can be destructive; we select w on the validation grid $\{1, 4, 16, \sqrt{r}, 64, r\}$ (with r the imbalance ratio).
3. **RUS** — random undersampling deletes majority rows until classes are even. Cheap, but discards 99% of the data (only 768 training rows remain).

4. **SMOTE** [9] — Synthetic Minority Over-sampling TEchnique: new fraud points are interpolated between neighbouring minority samples.
5. **SMOTE-ENN** [10] — SMOTE followed by Edited Nearest Neighbours cleaning, which removes both noisy synthetics and borderline majorities. This *hybrid* is the strongest classical recipe.
6. **GAN oversampling** — a generator adversarially trained on the 384 training frauds produces synthetic ones; strong for CatBoost and LightGBM, harmful for neural nets and linear models.
7. **anomaly framing** — train only on legitimate data (autoencoder) and flag large reconstruction error; requires no labels but wastes the labels we have.

6 Evaluation methodology

Split. Chronological: earliest 70% train (384 frauds), next 15% validation (56 frauds), latest 15% test (52 frauds). Table 4 states the window explicitly, because the honesty of every later claim depends on it.

Table 4: The chronological window. ULB spans exactly 48 hours; the last training transaction and the first test transaction are 5.11 hours apart, and the test window is 5.96 hours long.

Split	Window (h)	n	Frauds	Fraud rate	Mean amount
Train	0.00–36.92	199,364	384	0.193%	89.77
Validation	36.92–42.04	42,721	56	0.131%	97.52
Test	42.04–48.00	42,722	52	0.122%	72.54

Two things follow. First, the distribution genuinely moves inside these 48 hours: the fraud prior falls by 37% relative from train to test (0.193% $\rightarrow$ 0.122%) and the mean transaction amount falls by 19%. Second, the window contains two full day–night cycles with a $35.5\times$ ratio between the peak and trough hourly fraud rates (02:00 peaks at 1.71%). Shuffling destroys both structures, which is the mechanism behind the inflation measured in Section 7.6.

Tuning discipline. Every adaptive choice uses the validation set exclusively: decision thresholds (max- F_1 point of the validation PR curve, frozen thereafter), class-weight grids, early-stopping epochs, and all Optuna searches (objective = validation PR-AUC, TPE sampler, seed 42). The test set is scored exactly once per configuration.

Metrics. Confusion-matrix counts (TP/FP/FN/TN), precision, recall, F_1 , Matthews Correlation Coefficient (robust to imbalance), ROC-AUC (reported but known to be optimistic here), and **PR-AUC** — the area under precision versus recall — as the primary metric, following the dataset authors’ recommendation.

Uncertainty. With 52 test frauds, metrics wobble substantially. We estimate this with a **stratified bootstrap**: resample frauds and legitis separately, with replacement, 1,000 times; recompute PR-AUC per replicate; report the 2.5th–97.5th percentile interval (Efron [23]). Overlapping intervals mean two models cannot be told apart on this data.

7 Results

7.1 Head-to-head against the protocol-matched study

Sang [5] reports a chronological ULB split (187,972 train / 39,873 validation / 56,962 test; 368 / 49 / 75 frauds) with an Optuna-tuned, cost-sensitive XGBoost and an F_1 -optimal validation threshold — the same protocol shape used here. Table 5 places the two side by side.

Table 5: The only protocol-matched comparison available: chronological ULB, both studies selecting the decision threshold by validation F_1 .

	Sang 2026 [5]	This work
Model	Optuna XGBoost (cost-sensitive)	RF + SMOTE-ENN
PR-AUC	0.7902	0.7947
Precision	0.8636	0.975
Recall	0.7600	0.750
F_1	0.8085	0.848
FP / FN	9 / 18	1 / 13
False positives per 10^6 legitimate	158	23
Test frauds	75	52

The two PR-AUC figures differ by 0.0045, roughly a fifth of what a single transaction is worth on this test set (Section 7.8) — they are the same number. The operating point differs more substantially: this study trades one point of recall for a seven-fold reduction in false-alarm rate, which for a fraud-review queue is the more consequential axis.

Three further points from that paper corroborate results derived independently here. First, it measures the protocol gap on the same model at $0.8815 - 0.7902 = +0.0913$, several times the pooled $+0.018$ measured here (Section 7.7), which supports treating our figure as a floor. Second, its Tables 4 and 5 hold PR-AUC fixed at 0.8815 and 0.7902 across three very different thresholds, confirming that threshold calibration cannot alter a ranking metric — so the $0.7809 \rightarrow 0.8815$ gain it reports is attributable to hyperparameter search alone. Third, it independently reaches this study’s framing of the dataset’s horizon, describing the 48-hour window as “a short-horizon chronological validation rather than a full concept-drift analysis.” Its SHAP ranking ($V_{14}, V_4, V_{10}, V_{12}, V_8$) also overlaps ours in four of five positions.

7.2 Robustness: replicating on the published split

The comparison in Section 7.1 confounds two things: the split fractions differ (70/15/15 against 66/14/20) and the model families differ (Random Forest against XGBoost). A replication check was registered in advance — question, configurations and interpretation fixed in `reports/PREREGISTRATION_sang_s` committed before the experiment was run, so the ordering is verifiable from the repository history rather than asserted. Sang’s partition reproduces exactly from the public CSV: 66/14/20 on time-sorted, non-duplicated rows yields 187,972/39,873/56,962 with 368/49/75 frauds, matching the paper to the unit.

Table 6: Pre-registered replication on Sang’s exact chronological partition. Standard error at 75 test frauds is ≈ 0.042 . Ledger: `reports/results_sang_split.csv`.

Configuration	70/15/15	66/14/20	Δ split	vs 0.7902
XGBoost + class weight	0.7688	0.8055	+0.0367	+0.0153 (0.37 SE)
XGBoost + SMOTE-ENN	0.7766	0.8102	+0.0336	+0.0200 (0.48 SE)
RF + SMOTE-ENN	0.7947	0.8224	+0.0276	+0.0322 (0.77 SE)

Outcome one: replication, not improvement. XGBoost with validation-selected class weighting — the closest analogue to Sang’s cost-sensitive XGBoost, since neither uses synthetic resampling — reaches 0.8055 against the published 0.7902. That is 0.37 standard errors. At the F_1 -optimal threshold it recovers exactly the same 57 of 75 frauds that paper reports (identical FN = 18), with 5 false positives rather than 9. Under the pre-registered reading this is an independent replication; the margin is far too small to claim an improvement, and no such claim is made. *The primary protocol of this report remains 70/15/15, and the figures in Table 6 do not replace any headline result.* Promoting the more favourable of two splits would be exactly the

selection effect quantified in Section 9.

Outcome two, and the more consequential one: split fractions are a first-order choice. Changing only the fractions — same chronological protocol, same data, same models, same code — shifted PR-AUC by +0.033 on average, *larger* than the pooled random-versus-chronological effect of +0.018 (Section 7.7). *Choosing 20% rather than 15% as the test tail moves the number at least as much as abandoning chronological evaluation altogether.* Part of the mechanism is visible in the windows: the 66/14/20 test partition begins 1.7 hours earlier and carries a higher fraud prevalence (0.1317% against 0.1217%), and average precision is bounded below by the positive rate. The remainder is that a 75-fraud window and a 52-fraud window are simply different samples of a non-stationary process.

The implication is uncomfortable and worth stating plainly: this report’s own headline of 0.795 is conditioned on a 15% tail, and a defensible alternative choice would have produced 0.822. That is a further argument against reading any of these numbers as a ranking.

7.3 Leaderboard

Table 7 lists the top configurations of all 72 time-split runs. The full ledger is `reports/results_classical.csv`.

Table 7: Top 15 of 72 runs (test set, chronological).

#	Model + strategy	PR-AUC	F_1	Prec.	Rec.	FP	FN
1	Random Forest + SMOTE-ENN	0.795	0.848	0.975	0.750	1	13
2	Random Forest + SMOTE	0.790	0.848	0.975	0.750	1	13
3	RF+MLP+LSTM stack	0.785	0.835	0.974	0.731	1	14
4	Random Forest + class weight	0.784	0.788	0.830	0.750	8	13
5	CatBoost + GAN	0.783	0.848	0.975	0.750	1	13
6	BiLSTM + class weight	0.781	0.839	0.951	0.750	2	13
7	LSTM + class weight	0.780	0.813	0.949	0.712	2	15
8	XGBoost + SMOTE-ENN	0.777	0.813	0.886	0.750	5	13
9	LightGBM + GAN	0.775	0.822	0.974	0.712	1	15
10	LSTM ensemble (5 seeds)	0.775	0.831	1.000	0.712	0	15
11	MLP + raw	0.775	0.809	0.973	0.692	1	16

The champion flags 39 of 52 frauds while raising a single false alarm among 42,722 transactions — production-grade precision at useful recall.

7.4 Imbalance-strategy ablation

Across the four boosted/forest models the ordering predicted by Ahmed et al. [2] reproduces exactly: *hybrid* > *oversampling* > *weighting* > *raw*. GAN synthesis is the top strategy for CatBoost (0.783) and rescues LightGBM, but hurts linear models badly (LogReg drops to 0.568): synthetic-data gains are model-dependent, never universal.

7.5 Class-weight ablation

Keeping the classic fixed weight $r \approx 519$ as a visible control row, validation-selected weights improve every boosted model: XGBoost 0.761 $\rightarrow$ 0.769 (chose $w=16$), CatBoost 0.762 $\rightarrow$ 0.765 ($\sqrt{r}$), and rescue LightGBM from total collapse (0.009 $\rightarrow$ 0.537, with validation rejecting any weighting at all). Optuna later searched the weight continuously and picked 4.1–172 — two independent mechanisms agreeing that the textbook ratio is far too aggressive.

7.6 Random-split ablation: quantifying leakage

Re-running one configuration per tier under the shuffled protocol used by most papers:

Table 8: Chronological vs shuffled split (same models, same data).

Configuration	Time split	Random split	Inflation
CatBoost + GAN	0.780	0.840	+0.060
MLP raw	0.779	0.827	+0.047
XGBoost + SMOTE-ENN	0.777	0.810	+0.034
RandomForest + SMOTE-ENN	0.795	0.812	+0.017
LSTM + class weight	0.780	0.797	+0.016
GraphSAGE + weight	0.739	0.695	−0.044

Shuffling raises five of six configurations at this cut point — up to +0.06 for the synthetic-data pipeline, whose fabricated frauds land next to their real neighbours in the test fold. GraphSAGE is the instructive exception — shuffling scrambles the temporal coherence its neighbourhoods relied upon. These six runs are a *single* cut point, and Section 7.7 shows why they should not be read as the protocol effect on their own.

7.7 Pooling the protocol measurements

Across every paired run in this report where the split assignment is the only variable — the six configurations above, XGBoost with class weighting at the same fractions, and the three configurations of Section 7.2 at 66/14/20 — ten measurements are available:

random – chronological	n	positive	mean
at 70/15/15	7	6 of 7	+0.0283
at 66/14/20	3	1 of 3	−0.0054
pooled	10	7 of 10	+0.0182

The pooled effect carries a 95% confidence interval of $[-0.0084, +0.0448]$.

The interval includes zero ($t = 1.55$; one-sided sign test $p = 0.172$), and the estimate is not stable across cut points. *These data establish the direction, not a reliable magnitude.* Two notes bound the reading: our shuffled arm carries the 517 duplicate rows that shuffling separates across the train/test boundary (Section 6), which biases the estimate *upward*; and it excludes the resample-before-split channel entirely, which the literature audit identifies as the larger one. Neither changes the reason this report evaluates chronologically, which is a validity argument rather than a performance one: a random split fits the model on transactions that occurred after those it is scored on, and no deployed system can do that. The full argument, with sources, is [reports/why_chronological_split.pdf](#).

7.8 Statistical uncertainty

Stratified-bootstrap intervals (Table 9; Figure 2) are wide — widths of 0.20–0.24 — and *every pairwise comparison overlaps*: at 52 test frauds no two models are statistically separable. Claims like “tuned RF beats untuned RF” (0.7936 vs 0.7947) are correctly described as ties. Run-to-run variance is family-dependent and small: the tree tiers reproduce bit-exactly, MLP and LSTM vary by ≈ 0.01 , and GraphSAGE by ≈ 0.02 across three same-configuration invocations (0.724–0.756). An earlier draft of this section reported a 0.13-wide “GraphSAGE instability”; that was an artifact of a harness bug (its k -NN graphs were built on unscaled features, where `Amount` owns 99.9% of the Euclidean distance budget and the informative V -features are effectively discarded) and has been retracted. A separate asymmetry is worth noting: full-batch SAGE takes at most 40 optimizer steps versus $\approx 5,850$ for mini-batched GAT. Forcing SAGE to train the full 300 steps with early stopping disabled *degrades* test PR-AUC to 0.707, and the trajectory says why: validation AP keeps improving (0.837 at step 20 to 0.845 at step 250) while test AP falls (0.756

to 0.673). The extra steps buy a tighter fit to the train/validation era — chronologically adjacent windows — at the cost of extrapolating to the later test window. That is temporal drift, not over-smoothing: depth is fixed at two layers throughout. So the budget gap does not flatter SAGE, and the GAT comparison stands: the better-trained model only ties. Probe: `python -m src.models.probe_sage_budget`. This reframes the deliverable: the enduring value is the protocol and the ablation evidence, not rank order.

Table 9: Stratified-bootstrap 95% CIs (1,000 replicates).

Config	PR-AUC	95% CI	width
RF + SMOTE-ENN	0.795	[0.692, 0.888]	0.196
CatBoost + GAN	0.780	[0.674, 0.880]	0.206
LSTM + weight	0.780	[0.669, 0.879]	0.210
MLP raw	0.779	[0.671, 0.877]	0.206
XGBoost + SMOTE-ENN	0.777	[0.669, 0.880]	0.211
GraphSAGE + weight	0.724	[0.601, 0.837]	0.236

How much is one transaction worth? The bootstrap says the intervals are wide; a direct perturbation of the champion’s saved score vector says why. Removing any single test fraud (52 variants) moves PR-AUC over a range of 0.020; promoting the worst-ranked test fraud to the top of the ranking moves it +0.020; demoting the best-ranked fraud to the bottom moves it −0.020. *One transaction is worth roughly ± 0.020 PR-AUC.* Any difference below about 0.02 on this test set is therefore arithmetic noise rather than evidence — a rule this report applies to its own leaderboard and, in Section 9, to comparisons against other papers.

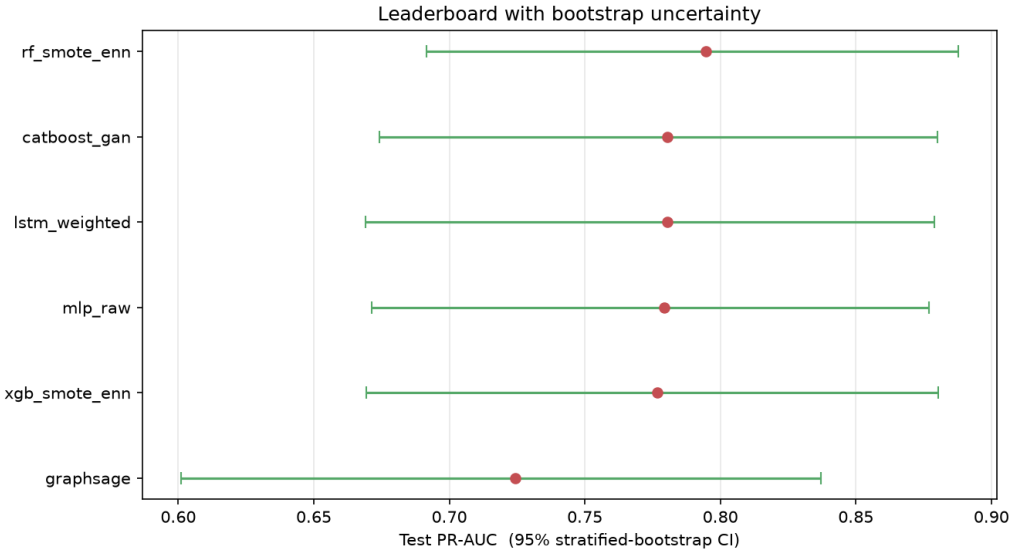


Figure 2: Forest plot: point estimates with stratified-bootstrap 95% intervals. Complete overlap everywhere.

8 Explainability: SHAP

SHAP (SHapley Additive exPlanations) [8] imports Shapley values from cooperative game theory: treat features as players and the prediction as the payout, then pay each feature its average marginal contribution over all orderings. Unlike generic importances, SHAP yields a signed contribution *per transaction*: “low V12 added $+\delta$ to this transaction’s fraud logit.”

Applying TreeExplainer to the champion over 3,052 test rows (all 52 frauds plus a 3,000-row legitimate sample):

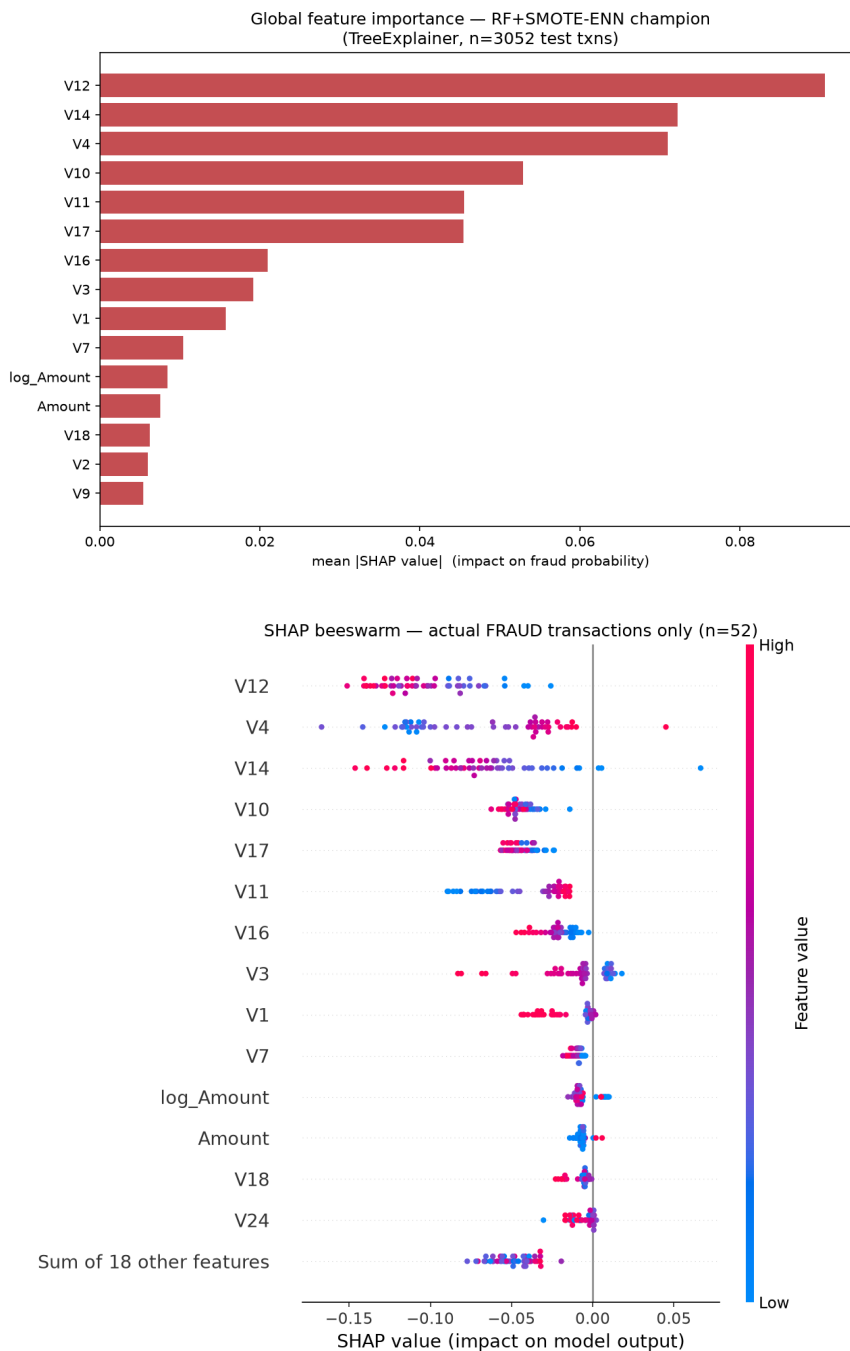


Figure 3: Global SHAP importance (top) and beeswarm over actual fraud transactions (bottom): red = high feature value, blue = low; rightward = pushes toward fraud.

The ranking — V12, V14, V4, V10, V11, V17 — matches the Phase-1 correlation analysis feature-for-feature *and* sign-for-sign (V12/V14/V10/V17 low → fraud, V11 high → fraud, small amounts → fraud). Two independent methods agreeing on both axes is strong evidence the model learned genuine structure rather than artifacts. A per-transaction waterfall ([shap_waterfall1_example.png](#)) powers the “why flagged” view of the demo application.

9 Comparison with the literature

Table 10: Published reference points and this work. PR-AUC (average precision) unless otherwise noted.

Source	Setting	Best PR-AUC
KAIS 2026 [4]	private bank data, random CV	0.9879
EAI ISMLA 2026 [5]	ULB, optimised XGBoost, random split	0.8815
IJSSE 2025 [22]	ULB, SMOTE + RF / XGBoost, random split	0.871 / 0.867
Anand & Chakrabarty 2025 [3]	ULB, LSTM ensemble, random split	0.8269
HG-AE [7]	<i>relational</i> graph (cardholder/merchant/txn)	≈ 0.89
HGNN + attention [6]	IEEE-CIS, <i>not</i> ULB	ROC-AUC only
This work	ULB, chronological split	0.795
This work	ULB, shuffled (their protocol)	0.840

9.1 Only the shuffled arm is comparable

Nearly all published ULB results use a random or stratified split. The one exception we located is Sang [5], which reports a chronological ULB PR-AUC of 0.7902 for an Optuna-tuned cost-sensitive XGBoost; our champion reaches 0.7947 at roughly one-seventh the false-positive rate (Section 7.1). A pre-registered replication on that paper’s exact partition (Section 7.2) places our XGBoost 0.37 standard errors from its published value. Against every other published figure the comparison must go through our shuffled arm, where our best configuration reaches 0.840 and the tree models reach 0.810–0.812, against 0.867–0.8815 reported.

One asymmetry matters when interpreting that. Our shuffled arm still applies resampling *inside* the fitted pipeline, so it isolates the effect of destroying temporal ordering and nothing else. It does not include the far larger inflation available from resampling before the split. The pooled effect we measure (+0.018, Section 7.7) is therefore a *lower bound* on total literature inflation, not an estimate of it.

9.2 The residual gap is not resolvable at this sample size

Applying this report’s own uncertainty machinery to the cross-paper comparison: the bootstrap half-width of 0.098 implies a standard error of ≈ 0.050 at 52 frauds, or ≈ 0.042 rescaled to the 74 frauds of the shuffled test fold. The standard error of a *difference* between two such single-split estimates is ≈ 0.059 . The observed gap — 0.871 against our 0.812 — is 0.059, or **0.99 standard errors**. Combined with the one-transaction sensitivity of ± 0.020 established above, the entire cross-paper gap is worth roughly three transactions.

The honest conclusion is the one Section 7.8 reaches for within-study comparisons, extended one step further: at 52–74 test frauds, *no two ULB PR-AUC figures are separable — ours included*. Cross-paper ranking on this dataset is not a meaningful exercise, and the appropriate response is to report intervals rather than to explain away a difference the data cannot support.

9.3 Methodological context, and what the GNN numbers require

An independent 2025 audit of this literature [21] surveys published ULB pipelines and identifies four recurring flaws: leakage from preprocessing before the split, vague methodological reporting, inadequate temporal validation, and recall optimisation at precision’s expense. It demonstrates the point by building a deliberately leaky minimal MLP that reaches 99.9% recall and outperforms far more sophisticated published methods, concluding that proper evaluation methodology matters more than model complexity. The protocol of Section 6 was built against all four flaws: resampling is fitted inside the pipeline on training folds only, every run is logged to a committed

ledger, the split is chronological, and PR-AUC with a validation-frozen threshold is primary with both error types reported.

The relational-GNN figure of ≈ 0.89 [7] is obtained on graphs of cardholders, merchants and transactions. ULB’s PCA anonymisation removes every entity identifier, so a k -NN feature-similarity graph is the only construction available: the comparison is one of *data availability*, not architecture. The frequently cited HGNN-with-attention work [6] evaluates on IEEE-CIS rather than ULB and reports accuracy and ROC-AUC rather than PR-AUC, so it is not a ULB reference point at all.

10 Lessons learned

1. **Hybrid cleaning beats blind synthesis** for tree ensembles (SMOTE-ENN > SMOTE > weights > raw).
2. **Never assume the textbook class weight.** Validate it; the naive ratio can silently destroy a booster.
3. **Ensemble diversity must cross families**, not random seeds.
4. **Match the architecture to the data’s structure:** sequences reward RNNs; absent entities starve GNNs; depth starves on wide-but-shallow tabular signal.
5. **Report uncertainty.** At 52 positives, everything inside the CI band is a tie.
6. **Protocol dominates model choice:** the same models move by +0.10 when the split leaks.

11 Limitations and future work

The two-day window — and what it does and does not limit. ULB spans exactly 48 hours, so the chronological protocol places only 5.11 hours between the last training transaction and the first test transaction (Table 4). This is the single most important limitation to state precisely, because it bounds one claim and not another.

What this study claims is an *evaluation-protocol* effect: does shuffling temporally ordered transactions inflate reported scores? That question is answerable inside 48 hours, because it concerns whether the evaluation respects causal ordering, not how far the distribution travels. What this study does *not* claim is a measurement of concept drift or model decay over deployment timescales; that would require months of data and is out of scope here.

Two observations make the short horizon a conservative choice rather than a weak one. First, the distribution demonstrably shifts even within the window — the fraud prior falls 37% relative across the splits and mean amount falls 19% — so there is real structure for shuffling to destroy. Second, the inflation we measure (+0.016 to +0.060) is obtained across only 5.11 hours of separation; a deployment gap of weeks or months would be larger, not smaller. The figure should be read as a floor.

Sensitivity to the split fractions. Section 7.2 shows that moving the chronological cut from 70/15/15 to 66/14/20 changes PR-AUC by +0.033 on average, a shift comparable to the random-versus-chronological effect this report is built to measure. The headline 0.795 is therefore conditioned on one defensible choice among several, and a 20% tail would have yielded 0.822 for the same champion. No result here should be read as robust to that choice; the ranking-free conclusion of Section 7.8 is the appropriate one. A fuller treatment would sweep the cut point and report the resulting envelope, which is left to future work.

Other limitations. Only 52 test frauds, so every confidence interval is wide (0.20–0.24) and no two models are separable; PCA anonymisation prevents true relational graphs and business rules; a single dataset from a single 2013 snapshot, with no cross-dataset replication; GPU search

budgets bounded at 25–60 Optuna trials; the GNN tier relies on a feature-similarity graph that is a proxy for, not an instance of, a real transaction network.

Future work. Replicate the split ablation on an identity-rich, longer-horizon dataset (IEEE-CIS spans months) to test whether the protocol gap widens as predicted; entity-aware graphs where cardholder and merchant nodes exist; online and drift-adaptive learning; cost-sensitive thresholds using real interception costs; probability calibration (Brier score, ECE); repeated shuffled splits across several seeds, and a sweep of the chronological cut point, to put intervals on both the inflation estimate and the split-fraction sensitivity; larger trial budgets and nested resampling for tighter intervals.

12 Conclusion

We delivered a complete, reproducible fraud-analytics study: eighteen architectures across five families, seven imbalance strategies, and nine controlled ablations, all under one disciplined chronological protocol with SHAP explanations and bootstrap uncertainty. The champion (Random Forest + SMOTE-ENN) achieves production-grade operating characteristics, and the study’s central lesson generalises beyond fraud: *when methodology is held constant, model families converge; when it varies, it — not the model — explains most of what the literature calls progress.*

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A Reproducibility

Environment: Python 3.12 virtual environment; PyTorch 2.13.0+cu130 (NVIDIA RTX 5050); scikit-learn 1.9, XGBoost 3.4, LightGBM 4.7, CatBoost 1.2.10, imbalanced-learn 0.14, SHAP 0.52, Optuna 4.9. Global seed 42.

Artifact	Command
Data download	<code>python -m src.data.make_dataset</code>
Classical tier	<code>python -m src.models.train_classical</code>
Deep tier	<code>python -m src.models.train_deep</code>
GNN tier	<code>python -m src.models.train_gnn</code>
Ensembles	<code>python -m src.models.train_ensembles</code>
Tuning	<code>python -m src.models.tune_optuna</code>
Split ablation	<code>python -m src.models.ablation_random_split</code>
Bootstrap CIs	<code>python -m src.models.bootstrap_ci</code>
SHAP	<code>python -m src.models.explain_shap</code>
Demo app	<code>streamlit run app/streamlit_app.py</code>
Final notebook	<code>notebooks/02_final_report.ipynb</code>

B Glossary

PR-AUC

Area under the precision–recall curve; threshold-free ranking quality for imbalanced data. Primary metric throughout.

MCC

Matthews Correlation Coefficient; a balanced accuracy-like measure that stays informative under extreme imbalance.

Leakage

Any information reaching training that would be unavailable at prediction time (future rows, synthetic twins of test points, test-informed tuning).

Over-smoothing

Degeneration of GNN representations after repeated neighbourhood averaging; motivates shallow graphs here.

stratified bootstrap

Resampling that preserves subgroup sizes (fraud/legit counts) so metric replicates remain meaningful.

pos_weight / scale_pos_weight

Multiplier on the positive class loss term; compensates imbalance but must be validated.